

ASIF SHAHRIAR

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RESEARCH INTERESTS

- Computer Security, AI Security, Vulnerability Detection, Adversarial ML
- Behavioural Security in LLMs and LLM-Agents

EDUCATION

Bangladesh University of Engineering & Technology (BUET) 2019 - 2024

B.Sc in Computer Science & Engineering

- CGPA: **3.85/4.0** [Scholaro US-equiv: **3.91/4.0**]
- CSE major-only CGPA: **3.91/4.0**
- **Dean's List** award for academic excellence

PEER-REVIEWED PUBLICATIONS

[1] Words Speak Louder Than Code: Investigating Cognitive Heuristics in LLM-Based Code Vulnerability Detection

Asif Shahriar, Hongyu Cai, Hadjer Benkraouda, Gang Wang, Z. Berkay Celik
IEEE Symposium on Security and Privacy (Oakland), 2027

[2] 5GPT: 5G Vulnerability Detection by Combining Zero-Shot Capabilities of GPT-4 With Domain Aware Strategies Through Prompt Engineering

A. Shahriar, S. J. Hisham, K. M. A. Rahman, R. Islam, M. S. Hossain, R. H. Hwang, and Y. D. Lin
IEEE Transactions on Information Forensics and Security (IEEE TIFS), 2025

[3] Inceptive Transformers: Enhancing Contextual Representations through Multi-Scale Feature Learning Across Domains and Languages

Asif Shahriar, Rifat Shahriyar, M Saifur Rahman
Proceedings of EMNLP 2025 (main)

[4] XLNet-CNN: Combining Global Context Understanding of XLNet with Local Context Capture through Convolution for Improved Multi-Label Text Classification

Asif Shahriar, Debojit Pandit, M Saifur Rahman
11th International Conference on Networking, Systems, and Security, 2024 (Best Paper Award)

PRE-PRINTS UNDER REVIEW

[1] A Survey on Agentic Security: Applications, Threats and Defenses

Asif Shahriar, Md Nafiu Rahman, Sadif Ahmed, Farig Sadeque, Md Rizwan Parvez
Under review in TMLR, 2026

ONGOING RESEARCH WORKS

Behavioural Side-Channel Inference on Browser-Use Agents

2026 - Ongoing

- Supervisor: **Dr. Rizwan Parvez (QCRI)**
- We investigate whether an attacker who only serves ordinary page content can infer a secret the agent holds from an earlier site, such as which bank it is signed into, purely from which link it clicks
- We build SIDE-BENCH, 20 scenarios and 100 tasks derived from the COSI state taxonomy
- Across 8,190 sessions and five backbones, 14 of 20 scenarios leak on every model, with mean leak rates of 0.53 to 0.66 per model and above 0.90 on the strongest scenarios

Compositional Bias in Generative Vision Language Models

2025 - Ongoing

- Supervisors: **Dr. M Saifur Rahman** and **Dr. Shubhra Kanti Karmaker (UCF)**
- In this work we introduce and define the concept of **compositional bias**, where VLMs use photographic aspects such as framing, focus, posture, and setting, to subtly reinforce stereotypes – making some groups visually “important” and others peripheral, even in seemingly diverse images.
- We identify TWO pillars of compositional bias – AGENCY (who is the primary actor) and FOCUS (when asked to choose a focal character, who does the model choose), and develop a new suite of metrics
- By analyzing over 1600 images across 45 scenarios, we have found that male characters get 2.3x more agency than females, non-white character gets less than 1% agency, and white characters dominate focal roles in most cases

TEACHING EXPERIENCE

Department of CSE, BRAC University

2024 – Current

Full-time Lecturer

[faculty-webpage](#)

- Artificial Intelligence (Theory and Sessional)
- Data Communications (Theory)
- Data Structures and Algorithms (Sessional)

PROFESSIONAL SERVICE

- **Reviewer**, EMNLP 2026
- **Reviewer**, IEEE Transactions on Cybernetics
- **Reviewer**, Cybersecurity (Springer)

TECHNICAL SKILLS

Languages: Python • Java • Javascript • C • C++ • $\text{\LaTeX}$ • Assembly

LLMs & Agents: RAG • fine-tuning • in-context learning • prompt engineering • tool-using agents • browser-use agents • coding agents • agentic security evaluation

Deep-Learning: PyTorch • TensorFlow • Neural Networks (ResNet, InceptionNet, ViT, RNN, LSTM) • Transformers • Language models

Systems & Tooling: Linux • Docker • Git • OracleDB • PostgreSQL

Networking & Security: 5G • LTE • Cryptography • EDR • NS2 • Open5GS • UERANSIM • 5GReplay • Scapy • CPPCheck • WireShark

HONOURS, AWARDS & GRANTS

- **Best paper award** in 11th International Conference on Networking, Systems, and Security (NSysS 2024)
- **Dean’s List award** for academic excellence in undergrad
- Full funding from ACL to present paper in the main conference of **EMNLP 2025**
- **Talent-pool scholarship** in Higher Secondary (**10th** in Dhaka Board)
- **Talent-pool scholarship** in Secondary (**26th** in Dhaka Board)

REFERENCES

- **Dr. Md. Shohrab Hossain**, *undergrad thesis supervisor, co-author*
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- **Dr. Z. Berkay Celik**, *research supervisor, co-author*
Associate Professor, Department of Computer Science, Purdue University
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- **Dr. Gang Wang**, *research supervisor, co-author*
Associate Professor, Computer Science, University of Illinois Urbana-Champaign (UIUC)
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